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After reading Ryall - 2009 - Causal Ambiguity.pdf , I think the conceptual boundary is much sharper.

The first paper should not present itself as a general theory of “Bayesianism under unawareness.” That literature already models expanding state spaces and belief revision—most notably reverse Bayesianism. The novel problem is the transition that those models leave unexplained:

$$\text{experience} \rightarrow \text{recognized causal gap} \rightarrow \text{question} \rightarrow \text{candidate insight} \rightarrow \text{new prior} \rightarrow \text{Bayesian}$$

In one sentence: **Bayes revises beliefs within a theory; insight changes the language in which theories can be believed.**

Where the 2009 paper leaves us

The causal-ambiguity model already handles two important learning problems:

Problem	What is fixed	What is learned
Parameter learning	Variables and DAG	Conditional probabilities
Structural learning	Variable vocabulary	Which DAG generated experience

The entrant knows all activity variables and all possible operating structures. Passive experience narrows these to an observational-equivalence class; experimentation can sometimes discriminate among the survivors.

That is an ideal benchmark for the new paper. But it is not yet unawareness: the correct graph is already in the entrant’s hypothesis space. The new paper should add a third level:

Problem	What is missing
Conceptual expansion	A variable, distinction, mechanism, action, or rival-theory state needed to formulate the appropriate causal model

Bayesian causal discovery and active intervention design remain in the first two rows: they rank experiments over a represented causal-model class. For example, standard active causal learning begins with known variables and a Markov-equivalence class. He and Geng ↗ provide a clean benchmark of that kind.

The key distinction: closed versus open questions

A **closed causal question** comes with conceived answers:

> Does $X \rightarrow Y$, or does $Y \rightarrow X$?

That is uncertainty, not unawareness. It can be represented as a partition of the current model set and handled by Bayesian experimental design.

An **open causal question** represents an explanandum and a standard of resolution without representing the answer:

> What accounts for the instability in Y?

> What unmodeled difference makes the rival react differently to apparently identical actions?

This is the important object. I would call it a **causal socket**: the manager knows where her theory is incomplete and what an adequate completion must accomplish, but she cannot yet describe the completion.

Formally, something like

$$q = (r_q, \tau_q, \mathcal{E}_q, R_q),$$

where:

- r_q is the localized defect or explanandum;
- τ_q is a coarse repair type—missing cause, omitted distinction, mediator, reversed influence, rival-theory mechanism;
- $\mathcal{E}_q$ is the feasible inquiry or intervention technology;
- R_q is the criterion an adequate answer must satisfy.

Crucially, q contains no semantic list of answers.

That gives us a substantive interpretation of your "human superpower": **forming a question is itself an epistemic achievement**. Experience has converted an unknown unknown into a localized, structured unknown.

How experience could generate questions

Let the manager's current language L_t induce a representation—or partition—of experienced episodes. Two episodes may be remembered but currently classified as "the same kind of situation."

Now define two histories to be causally equivalent when they imply the same future outcome distributions under every feasible intervention, including strategic responses:

$$h \sim^* h' \quad \Longleftrightarrow \quad P(Y^+ \mid do(a), h) = P(Y^+ \mid do(a), h') \quad \text{for every feasible } a.$$

A causal anomaly occurs when the manager's current concepts pool histories that experience increasingly suggests are not causally equivalent. Examples include:

- repeated residual dependence;
- failure of a mechanism to remain invariant across contexts;
- different rival responses to apparently equivalent moves;
- several observationally equivalent structures that imply different optimal interventions.

The resulting question is not drawn from an infinite urn. It is generated locally:

> What distinction inside this pooled category explains the contrasting responses?

This is where the 2009 machinery becomes very useful. It supplies the **closed-question frontier**: experience produces an equivalence class, and the unresolved edges generate possible experiments. The new machinery begins when no adequately simple model in the current vocabulary provides a stable account—so the agent must seek a new distinction rather than merely orient a known edge.

A technical qualification matters: with one passive observational distribution, a sufficiently dense DAG over existing variables can usually fit almost anything. To identify language inadequacy, the paper will need some combination of multiple environments, interventions, modularity, invariance, sparsity, or complexity restrictions. That is not a nuisance; it is probably where the interesting theorem lives.

Formalizing “this problem probably has an answer”

An answerable question needs three properties:

1. An objective causal completion exists.
2. It is reachable using the manager’s constructive repertoire—analogies, concepts, decompositions, and permissible representational operations.
3. Available inquiry can discriminate it from relevant alternatives.

Thus:

$$A_{it}(q) = 1$$

if there exists a constructible causal refinement ι that repairs the targeted defect and is identifiable using feasible inquiry.

The manager need not know ι. She can instead possess a **meta-prior over answerability**:

$$p_{it}^{Q}(q) = \Pr(A_{it}(q) = 1 \mid \phi_{it}(q)) \,,$$

where $\phi_{it}(q)$ contains features such as:

- recurrence and localization of the anomaly;
- cross-context regularity;
- proximity to previously solved problems;
- availability of discriminating interventions;
- apparent simplicity or compressibility;
- accessibility from the manager’s conceptual repertoire.

This is a prior over whether working on a problem will yield an intelligible causal completion—not a prior over unimagined answers. That seems to me the cleanest formal interpretation of the superpower you are describing.

But it only becomes theoretically interesting if the diagnostic force of those features is derived. Under a sparse, modular causal world, for example, a persistent localized invariance failure might genuinely imply that the missing mechanism lies in a small causal neighborhood. Directed inquiry would then outperform undirected representation search. Merely assuming that “promising-looking questions tend to have answers” would reproduce the baked-in problem.

Ranking questions

The ranking should combine answerability with strategic leverage, not merely information or entropy.

A parsimonious index would be

$$I_t(q) = p_t^Q(q) B_t(q) - c_t(q),$$

where $B_t(q)$ is the robust decision value of resolving the gap.

Here the C^+ agent supplies useful machinery. Its frontier-refinement logic can ask:

> Could resolving this causal gap overturn the value of the currently preferred plan?

The bounds must be subjective, coarse, and potentially wrong. If the agent knows exact objective payoff bounds behind every unconceived possibility, unawareness has slipped out of the model.

An even safer approach would first establish a partial ordering: q dominates q' when it exhibits at least as much anomaly pressure, at least as much possible action separation, at least as much resolvability, and no greater cost. A scalar index can then be introduced only where needed.

This also produces an important TBV result: **the most informative question need not be the most strategically valuable question**. A high-entropy causal ambiguity is irrelevant when every answer supports the same plan. A narrow ambiguity can be crucial when its answers lie on opposite sides of an action threshold.

What exactly is an insight?

I would not define insight as a posterior crossing a threshold. That would collapse insight into judgment and lose the Lonergan structure.

An insight should be a candidate language expansion

$$\iota : L_t \hookrightarrow L_{t+1}$$

satisfying four conditions:

1. **Representational novelty:** the new distinction or mechanism was not definable in L_t.
2. **Empirical organization:** it materially improves predictive adequacy, compression, or causal invariance for the experience targeted by the question.
3. **Causal content:** it creates at least one new interventional or counterfactual implication.
4. **Minimality:** no strictly smaller expansion achieves the same causal repair.

A **strategic insight** additionally changes the optimal action for some relevant region of beliefs.

Most importantly, an insight need not be true. It supplies a candidate intelligibility. Formulation draws out its implications; judgment tests whether it is so. False insights, failed inquiry, multiple adequate explanations, and unreachable truths must all be possible. Otherwise truth has again been built into the expansion operator.

The Bayesian connection

Before insight, the manager has a prior π_t only over models expressible in L_t. Ordinary conditioning cannot make an event outside that language measurable. Once insight creates L_{t+1}, beliefs must be extended to the richer model space.

If ρ projects richer models back onto their old descriptions, a bridge prior might take the form

$$\pi^0_{t+1}(m') = \pi_t(\rho(m'))\,\kappa_\iota(m' \mid \rho(m'))\,.$$

This preserves the old marginal beliefs while κ_ι allocates credibility among newly expressible refinements. Complexity, analogy, source-question features, and previously learned inquiry competence might discipline that kernel.

But Bayes does not determine κ_ι. There are generally many expanded priors consistent with the old one. That nonuniqueness is the boundary between insight and Bayesian judgment. Karni and Vierø's reverse Bayesianism ↗ provides a coherence principle for growing awareness, but preserving old relative odds does not itself explain which insight appears or how probability is distributed within the newly opened possibilities. Later work explicitly notes that reverse Bayesianism alone does not fully determine the expanded distribution. Chakravarty, Kelsey, and Teitelbaum ↗

There is also a discovery-data problem. If anomalous experience generated the insight, it cannot simply be reused as though the expanded theory had been specified beforehand. The clean treatment for paper one is:

- use one history to generate the question and candidate insight;
- construct a reference prior on the expanded space;
- use prospective or held-out experience for Bayesian judgment.

That maps beautifully onto Lonergan's separation of insight from reflective judgment.

Making the game-theoretic element essential

The objective environment should be a structural causal game whose state includes rivals' theory states:

$$\theta_{jt} = (L_{jt}, M_{jt}, \pi_{jt}), \qquad a_{jt} = \sigma_j(h_{jt};\theta_{jt}).$$

Thus rival theory causes rival action, and the focal manager may initially omit that causal mechanism:

$$a_i \to \text{rival observation} \to \text{rival theory} \to a_j \to u_i.$$

Structural causal games already provide machinery for combining decisions, interventions, counterfactuals, and strategic relevance. Hammond et al. ↗

For tractability, only the focal manager should undergo endogenous awareness expansion in the first paper. Give the rival a finite hidden theory type and a specified response/update rule. That is enough to generate the genuinely strategic result:

> Because current theories determine actions, and actions determine experience, a coarse or false causal theory can suppress the very experiences that would generate the question needed to overturn it.

This is an **awareness trap**, stronger than ordinary self-confirming belief. Schipper distinguishes learning—which removes conceived possibilities—from discovery—which adds previously unconceived possibilities; his discovery processes provide a useful comparison, but they do not explain how experience selects a substantive causal question. Schipper ↗

The paper I would write

Working title:

> **Questions Before Priors: Directed Causal Inquiry under Unawareness**

The model would contain:

- one focal strategist and one strategically responsive rival;
- an objective causal game;
- the strategist's coarse causal language;
- experience-generated causal contrasts and invariance failures;
- open questions as localized causal sockets;
- a learned meta-prior over question answerability;
- C^+-style valuation of decision leverage;
- insight as minimal causal-language refinement;
- prior formation and Bayesian judgment after insight.

The central results would be:

1. Bayesian updating cannot generate a genuinely new causal distinction, and prior extension after insight is nonunique.
2. Under causal modularity and sparsity, experience localizes a payoff-relevant missing mechanism, making inquiry genuinely directed.
3. Strategic question value depends on action separation, not information quantity.
4. Endogenous play can sustain self-confirming unawareness by preventing the experience that would generate the decisive question.

I would defer full C^{Plan}, reciprocal insight by every player, collective intentionality, and value capture. Those are natural sequels. Here C^+ contributes the selection logic; the 2009 paper contributes the full-awareness causal benchmark; Lonergan supplies the separation of question, insight, and judgment; and game theory supplies endogenous experience and awareness traps.

The next move should be very concrete: construct the smallest two-player example in which an omitted rival-theory variable creates a localized causal anomaly, then see whether we can prove the localization and awareness-trap results without ever handing the manager the answer. If that toy model works, this is no longer merely an agenda—it is a paper.